Bocconi

CORPORATE FINANCE 30017

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Return Data Analysis

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Agenda

DATA ANALYSIS: LEARNING FROM PAST

EXPECTED RETURN

RETURN VARIANCE

RETURN DISTRIBUTION MODELLING (PRIMER)

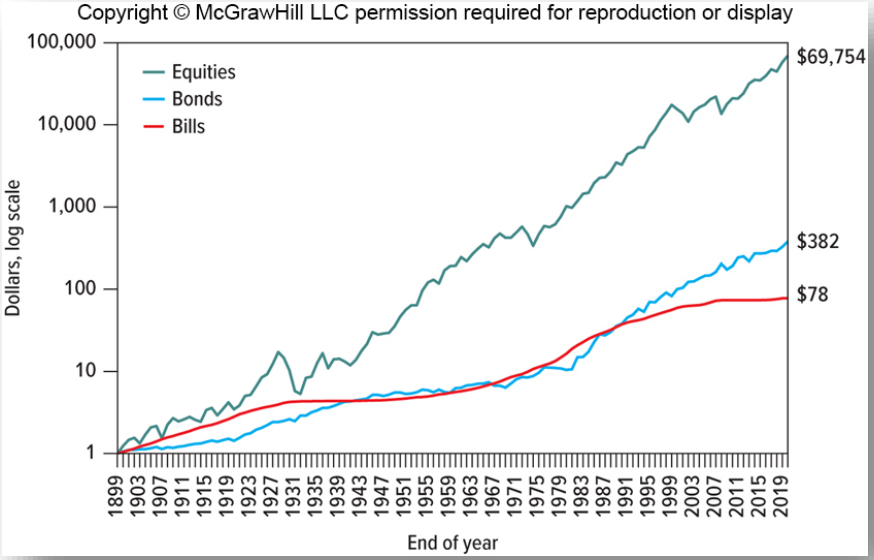
DATA ANALYSIS: LEARNING FROM PAST

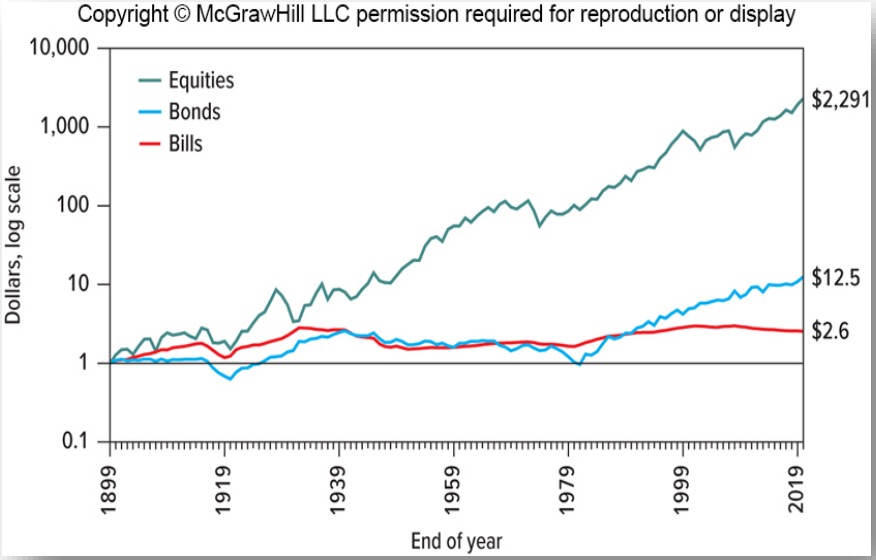
Holding Period Return: Definition

$$r_{0,H} \equiv \frac{\sum_{h=0}^{H} CF_h + (P_H - P_0)}{P_0}$$

- Total (e.g., multi-year) return, calculated as sum of interim cash flows and price change relative to initial price

Long-Term Holding Period Returns





— *Did bond investors make mistakes for decades in a row…?*

Learning from the past

- When we look into the future, we (have to) make **predictions**
  - What are possible scenarios?
  - What is the probability of each of those scenarios?
  - That is, we need to **characterize ("model") the distribution of future scenarios**

- Ultimately, there is no other way than inferring those predictions from experience (past)
  - Data characterizes past quantitatively: applicability of math tools (statistics)
  - **Can use data to estimate (moments of) the distribution of future scenarios**
  - *What if the underlying distribution changes over time…? (Nemesis)*

- This logic applies to all prediction, **including predictions about returns**

EXPECTED RETURN

Expected Return

- **Expected return** $\boldsymbol{E}[r]$ of some asset is probability-weighted average of all possible returns
  - Probability-weighting across all possible scenarios ("states" s)

$$\boldsymbol{E}[r] = \sum_s p_s r_s$$

- where
  - s are possible scenarios ("states of the world"),
  - p_s is the probability of a given state, and
  - r_s is the return in that state.

Estimating the Expected Return

- Consider the arithmetic average of past returns

$$\bar{r} = \frac{1}{T}\sum_{t}^{T} r_t$$

- where
  - t are past points in time $t \in \{0,1,2,\dots,T\}$
  - r_t is the return observed at given point in time

- Can be used to estimate expected return $\boldsymbol{E}[r]$
  - Assume: return observations are statistically independent from each other
  - **Law of large numbers** implies arithmetic average will be close to expected value with many observations

$$\bar{r} \xrightarrow{T\to\infty} \boldsymbol{E}[r]$$

Side note: Arithmetic v. Geometric Average

— **Arithmetic average** is powerful (and thus ubiquitous) in statistics

- But can be misleading in financial applications:

$$V_0 = 100, V_1 = 110, V_2 = 100$$

$$r_{0,2} = 0\%$$

$$r_1 = 10.0\%, \qquad r_2 = -9.1\%$$

$$\bar{r} = 0.5\%$$

- Use **geometric average** $\tilde{r}$ to obtain return that compounds to holding period return:

$$(1+\tilde{r})^T = \left(1 + r_{0,T}\right)$$

$$\tilde{r} = \sqrt[T]{1 + r_{0,T}} - 1$$

RETURN VARIANCE

Variance and Standard Deviation

- To quantify uncertainty around possible outcomes, we most frequently use variance
  - or standard deviation (square root of variance)
- **Variance** is the expected squared deviation from a variables expected value

$$Var(r) = \boldsymbol{E}[(r - \boldsymbol{E}[r])^2]$$

- Variance is hard to interpret, hence, popularity of standard deviation
  - Expected deviation from the mean
  - *"If you draw a random return, how far to you expect it to be from its mean?"*

$$SD(r) = \sqrt{Var(r)}$$

Estimating Return Variances

— Empirical variance

$$\sigma^2 = \frac{1}{T-1}\sum_{t=0}^{T}(r-\bar{r})^2$$

- will converge to theoretical variance (under same assumptions of law of large numbers):

$$\sigma^2 \xrightarrow{T\to\infty} Var[r]$$

- Adjusted denominator $\frac{1}{T-1}$
  - "degrees of freedom adjustment" since we implicitly estimate $\boldsymbol{E}[r]$
    - *Think of a sample of size 1…*
  - In Excel: use VAR.S() to get adjustment; VAR.P() will instead calculate $\frac{1}{T}\sum_{t=0}^{T}(x-\bar{x})^2$

Volatility vs. Risk

- Volatility is not necessarily a reflection risk
  - Think about a pendulum….
- However, if observations are statistically independent, volatility corresponds to variance (risk)

Covariance

- Covariance is a measure of *co-movement*, or *relatedness*.
  - It is defined as the expected product of deviations from means:

$$Cov(r_i, r_j) = \boldsymbol{E}\big[(r_i - \boldsymbol{E}[r_i])(r_j - \boldsymbol{E}[j])\big]$$

  - *What happens if* $i = j$*?*

- Estimate with the sample counterpart, analogous to variance (mind degree of freedom adjustment)

Correlation

— Correlation between two returns is **covariance normalized** by product of SDs

$$\rho(r_i, r_j) = \frac{Cov(r_i, r_j)}{SD(r_i)SD(r_j)} \in [-1,1]$$

— Intuition: Normalization bounds correlation between -1 and +1.

- $\rho(r_i, r_j) = 1$: Perfect positive (linear) relationship
- $\rho(r_i, r_j) = -1$: Perfect negative (linear) relationship

— Note, trivially: $Cov(r_i, r_j) = \rho(r_i, r_j)SD(r_i)SD(r_j)$

Relevant Functions in Excel

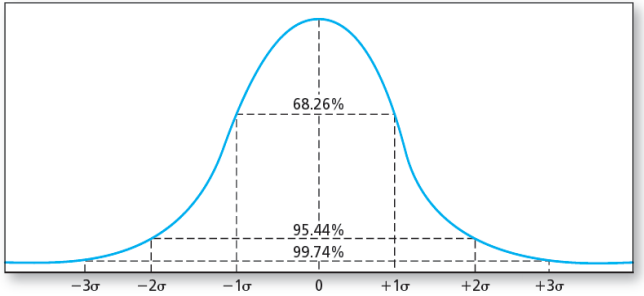
- Arithmetic average `AVERAGE`
- Variance - `VAR.S` (for a sample), `VAR.P` (for the population)
- Standard deviation - `STDEV.S` (for a sample), `STDEV.P` (for the population)
- Covariance - `COVARIANCE.P` (for a sample), `COVARIANCE.P` (for the population)
- Correlation - `CORREL`
- Sum of the products - `SUMPRODUCT`

RETURN DISTRIBUTION MODELLING (PRIMER)

Is this normal?

— Finance theory can provide more useful/elegant predictions when returns are normally distributed

$$r \sim \mathcal{N}(\mu, \sigma)$$



Density function: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

- Full distribution parameterized by two moments:
  - μ equal to (estimable by) expected value
  - σ equal to (estimable by) standard deviation
- Other nice features
  - If security returns are normal, portfolio returns will be normal as well
  - The dependence of returns across securities can be fully summarized by pairwise covariance

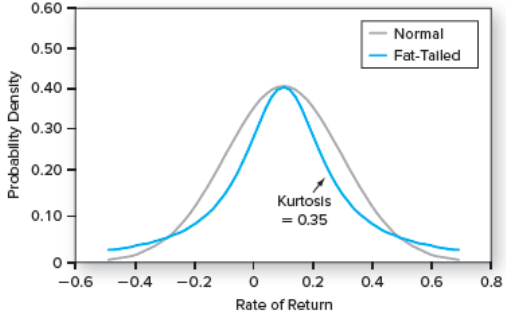
Deviations from Normality

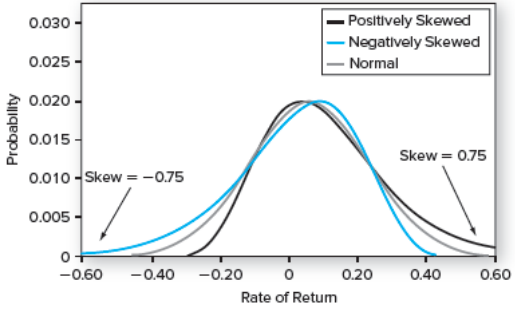
- Normal distribution generally good approximation of returns – but there are caveats
  - "Fat tails": extreme returns occur more frequently than normality predicts
  - "Asymmetry": returns are more (or less) often above than below the mean
- Detecting deviations from normality: Check Kurtosis and Skewness

$$Kurtosis = \frac{\boldsymbol{E}[(r - \boldsymbol{E}[r])^4]}{SD(r)^4}$$

$$Skewness = \frac{\boldsymbol{E}[(r - \boldsymbol{E}[r])^3]}{SD(r)^3}$$

  - Under normality, kurtosis should be (close to) 3, skewness should be (close to) 0





Testing the Model

In class exercise:

- Download NVIDIA's and McDonald's monthly stock returns for 1990-2020 from WRDS
  - CRSP, Stock/Security Files/ Monthly Stock File
  - Set data range, Ticker (NVDA, MCD)
  - Select: ticker, ret (permno selected by default)

- Can NVIDIA's or McDonalds returns be modelled by normal distribution?
  - Create histogram of monthly returns
  - Overlay normal (density) distribution with data-based mean and SD
  - Check Kurtosis and Skewness
  - *Can AI do it for you?*